

Node.js & Express Bootcamp – Module 3 Exercise

Hands-on Exercise No. 3

Course: Node.js & Express Bootcamp

Module: Authentication

Total Marks: 10

Due Date: 26/05/2026

Instructions:

- Prepare your solution in MS Word
 - Submit in .doc or .docx format
 - Late or copied submissions will not be accepted
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Learning Outcome:

- Implement user authentication in Express
 - Use JWT (JSON Web Tokens) for secure login
 - Protect routes using middleware
 - Handle registration and login flows
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Problem Statement:

You are working as a Backend Developer for an Online Learning Platform. The client requires a secure authentication system for students and instructors. Your task is to implement JWT-based authentication in an Express application.

Task Requirements:

- Create a new Express project
- Install required packages:
 - express
 - jsonwebtoken
 - bcrypt (optional for password hashing)
- Implement the following routes:
 - POST /register → Register a new user with email and password
 - POST /login → Authenticate user and return JWT token
 - GET /profile → Protected route that returns user details only with valid token
- Add middleware to verify JWT tokens before accessing protected routes

- Store user data using:
 - In-memory array OR MongoDB (optional)
 - Test API using Postman:
 - Register user
 - Login and get token
 - Access protected profile route
 - Try accessing without token (should fail)
 - Take screenshots of:
 - Registration request/response
 - Login and token generation
 - Protected route access
 - Failed access without token
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Submission Guidelines:

- Upload your .docx file on LMS
- Include:
 - Code snippets
 - Postman screenshots
 - JWT token verification output
- Add short explanation for each API flow